# PropNet: Detecting Misinformation via Hybrid Graph Attention and Structural Propagation Motifs

Srivathsa H. Honyal, Ashmit Dhown, Aaryan Gupta, and Sachin P.
*Department of Computer Science*
*BITS Pilani, Dubai Campus*
Dubai, UAE
{f20230198, f20230205, f20230360, f20230208}@dubai.bits-pilani.ac.in

***Abstract*—The proliferation of misinformation across online social networks poses a significant threat to public discourse and societal well-being. Existing automated detection systems predominantly rely on natural language processing (NLP) pipelines that analyze textual content in isolation—an approach increasingly brittle against adversarially crafted narratives engineered to evade lexical detection. In this paper, we present PropNet, a hybrid Graph Neural Network (GNN) framework that reframes misinformation detection as a graph classification problem by jointly modeling article claims and their propagation dynamics. PropNet's central architectural contribution, HybridGATWithAttention, fuses 768-dimensional DistilBERT semantic embeddings with a hand-crafted 7-dimensional network motif vector via a learned gated attention mechanism. Evaluated on the pooled UPFD benchmark comprising PolitiFact and GossipCop corpora, PropNet achieves a ROC-AUC of $0.989 \pm 0.004$. Integrity is verified through a rigorous 5-seed ablation study and label-shuffle sanity checks, demonstrating that structural topology constitutes a powerful, manipulation-resistant signal for automated misinformation detection.**

***Index Terms*—Graph Neural Networks, Misinformation Detection, Graph Attention Networks (GAT), Multi-modal Fusion, Network Motifs, Social Network Analysis.**

## I. Introduction

The democratization of social media has transformed global discourse into an efficient vector for fabricated content, which now spreads at a scale that far outpaces manual fact-checking [1], [2]. These disinformation campaigns have severe consequences, from electoral interference to the erosion of public trust [3], [10]. Traditional detection systems treat this primarily as a text classification task, using NLP pipelines to analyze article content [4], [12]. However, these approaches rely on the fragile assumption that false content is linguistically distinct. In adversarial settings, actors can craft misinformation that is stylistically identical to legitimate journalism, rendering text-only classifiers fundamentally inadequate [13], [19].

In contrast to easily manipulated text, the structural topology of how a message spreads through a social network offers a more robust signal for detection. This propagation graph encodes the sequence of reshares and the underlying community structures, reflecting behavioral patterns that are extremely difficult for malicious actors to replicate at scale [5], [7].

Research indicates that genuine news typically diffuses organically across diverse communities, which often results in shallow cascade graphs. In contrast, misinformation frequently spreads through coordinated inauthentic behavior, creating tightly clustered echo chambers and star-shaped amplification networks [16], [22]. Forging the emergent structural signature of organic diffusion across thousands of independent interactions is socially and technically intractable.

Graph Neural Networks (GNNs) are the natural algorithmic substrate for this task [4], [18]. By operating directly over graph-structured data, GNNs learn topology-aware representations of propagation cascades. Graph Attention Networks (GATs) extend this by computing attention-weighted aggregations, allowing the model to focus on structurally anomalous subgraphs that serve as telltale signatures of coordinated behavior [11], [14].

Despite the promise of graph-based approaches, a persistent limitation in existing literature is the treatment of text and structure as mutually exclusive modalities. Purely structural approaches forfeit semantic signals, while purely textual approaches forfeit the robustness afforded by propagation topology [9], [23]. We address these limitations with *PropNet*, a hybrid GNN framework designed for principled multi-modal fusion. Our core architectural innovation, *HybridGATWithAttention*, processes each news propagation graph through parallel pipelines: a multi-layer GAT stack for semantic embeddings and a structural feature extractor for topological fingerprints. These are combined via a learned gated attention mechanism that adaptively weights each modality on a per-instance basis.

The primary contributions of this work are as follows:

- **HybridGATWithAttention Architecture:** A novel multi-modal GNN architecture that fuses DistilBERT semantic representations with structural motifs via an instance-adaptive gated attention mechanism [21].
- **Structural Motif Feature Engineering:** A principled 7-dimensional network motif descriptor capturing propagation topology across multiple scales, including triangle density, star-hub sizes, and cascade depth [16].
- **Rigorous Experimental Validation:** A comprehensive multi-seed evaluation protocol demonstrating state-of-the-art accuracy, achieving a final ROC-AUC score of $0.989 \pm 0.004$ on the unified benchmark.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset and preprocessing. Section IV details the PropNet architecture.

Section V presents experimental results, and Section VI concludes the paper.

## II. Related Work

### A. Content-Based Detection

Early automated approaches to fake news detection treated the problem as a variant of standard text classification. These systems extracted hand-crafted linguistic features, including n-gram distributions, punctuation patterns, and sentiment polarity scores, to train supervised classifiers. The underlying intuition was that fabricated content would exhibit statistically distinguishable linguistic signatures such as exaggerated sentiment or sensationalist vocabulary.

The introduction of deep learning marked a decisive shift in methodology. Yang et al. [4] proposed TI-CNN, a convolutional neural network architecture that learns latent textual and visual features from news content, demonstrating that learned representations outperform hand-engineered feature sets. Concurrently, Pérez-Rosas et al. [5] conducted a systematic evaluation of machine learning classifiers across multiple fake news datasets, establishing that while stylistic cues are informative, they are increasingly vulnerable to adversarial manipulation.

### B. Social Context and Propagation

Research has increasingly focused on the social context of information diffusion, motivated by the observation that false news propagates fundamentally differently than the truth. Vosoughi et al. [1] established that misinformation spreads significantly faster, deeper, and more broadly than truthful information across all categories of content. This phenomenon is often driven by coordinated inauthentic behavior. Shao et al. [8] demonstrated the critical role of social bots in the early stages of misinformation viralization, while Starbird et al. [9] characterized disinformation as a collaborative process involving strategic information operations.

### C. Graph-Based Learning

Graph Neural Networks (GNNs) have emerged as the standard algorithmic framework for modeling these propagation dynamics. Wu et al. [10] provided a comprehensive foundation for how GNNs learn representations from graph-structured data through neighborhood aggregation. The introduction of Graph Attention Networks (GAT) by Veličković et al. [11] further advanced the field by allowing models to adaptively weight the importance of different neighbors in a cascade. While these models capture local neighborhood structures, standard GNN implementations often fail to explicitly account for the specific network motifs, such as triangles or star-hubs, that serve as signatures of coordinated amplification.

### D. User Preferences and Multi-modal Fusion

The state of the art has moved toward integrating user-level characteristics with propagation topology. Dou et al. [14] introduced the User Preference-aware Fake News Detection (UPFD) framework, which incorporates user historical preferences to provide context for resharing behavior. Theoretical surveys by Qian et al. [12] and Zhou and Zafarani [13] emphasize that while multi-modal models are superior, the field still lacks efficient mechanisms for instance-adaptive fusion.

PropNet addresses this limitation via its *HybridGATWithAttention* architecture. Unlike models that rely on simple feature concatenation, our system introduces a learned gated attention mechanism. This allows the model to determine, on a per-sample basis, whether to prioritize the semantic representation derived from DistilBERT features or the structural representation derived from the 7-dimensional network motif vector. This design ensures that structural signals receive higher weight when textual content is designed to be adversarially neutral.

## III. Data Domain and Feature Engineering

This section describes the benchmark dataset on which PropNet is evaluated, the node feature representation used as input to the graph attention layers, and the structural motif engineering pipeline that constitutes one of PropNet's primary technical contributions. The design choices at each stage are motivated by the properties of social media propagation data and by the objective of constructing features that are both discriminatively powerful and robust against the data pathologies common in large-scale social graph corpora.

### A. The UPFD Benchmark Dataset

PropNet is trained and evaluated on the User Preference-aware Fake News Detection (UPFD) dataset introduced by Dou et al. [14], a graph-structured benchmark derived from the FakeNewsNet repository. UPFD provides pre-constructed propagation graphs for two distinct news domains: *PolitiFact* (614 graphs, political news verified by the PolitiFact fact-checking organization) and *GossipCop* (5,464 graphs, entertainment news evaluated by GossipCop). Each graph in the dataset represents the complete observed propagation cascade of a single news article on Twitter, yielding a binary graph classification problem in which the target label indicates whether the source article is real (0) or fake (1).

The graph topology is built by the following process. Each graph One source news is represented by one root node. posted on the Twitter platform, with every other node signifying users who Interacted with the material. Directed edges E are a path The number of retweets or shares of information. This formulation reframes misinformation detection as the classification of tree- The model is able to learn due to structured propagation cascades, from the structural dynamics of news diffusion rather than solely from article text.

### B. Node Feature Representation (DistilBERT)

While the primary focus of PropNet is structural propagation, the semantic content of the news claim provides the necessary foundation for the graph attention mechanism. To represent node-level features, we utilize pre-computed 768-dimensional embeddings generated by a DistilBERT model

[11]. DistilBERT is a distilled version of the BERT architecture that retains approximately 97% of the original model's performance while being significantly more computationally efficient. For each graph, the feature matrix $\mathbf{X} \in \mathbb{R}^{N \times 768}$ serves as the initial state for the GAT layers, where $N$ is the number of nodes in the cascade. These embeddings capture high-level semantic information about the users and the claim, which the subsequent GAT layers then aggregate according to the graph topology.

### C. Structural Motif Feature Engineering

The core technical contribution of our feature pipeline is the extraction of a 7-dimensional structural motif vector $\mathbf{M}$ for each graph. These features are designed to provide a compact, interpretable "topological fingerprint" that captures the structural anomalies characteristic of misinformation cascades. The seven features are defined as follows:

- **Triangle Count:** The total number of completed triangles in the graph, representing local clustering and the potential formation of "echo chambers" where users are densely interconnected.
- **Max Star Size:** The maximum degree of any single node in the graph. Large star motifs are often indicative of "broadcast" propagation patterns where a single influential node or bot accounts for the majority of the spread.
- **Average Clustering Coefficient:** The mean of the local clustering coefficients for all nodes, providing a measure of the overall "cliquishness" of the propagation network.
- **Graph Density:** The ratio of actual edges to the total possible edges in the graph, indicating the global connectivity saturation of the cascade.
- **Node Count:** The total number of participants in the cascade, serving as a proxy for the absolute reach of the news item.
- **Root Out-degree Fraction:** The ratio of edges originating from the root node to the total number of edges. High values indicate a flat, broadcast-heavy spread, while low values suggest an organic, multi-generational diffusion.
- **Depth Proxy:** A measure of the maximum distance from the root node to any leaf node, capturing the vertical elongation of the propagation tree.

### D. Preprocessing and Normalization

Propagation data in social networks typically follows a power-law distribution, where a small number of cascades exhibit extremely high node counts or star sizes while the majority remain small. To ensure numerical stability and prevent high-magnitude features from dominating the gradient updates, we apply a log-transformation to all count-based features:

$$x_{motif} = \log(x + 1) \tag{1}$$

This transformation is standard practice for count data in social network analysis [16] and compresses the dynamic range of each feature to a scale that is compatible with the remaining three continuous motif features (Average Clustering Coefficient, Graph Density, and Root Out-degree Fraction), which are naturally bounded in $[0, 1]$.

Following log-transformation, all seven motif features are standardized via a z-score scaler fitted exclusively on the training split and applied without refitting to the validation and test splits. This train-only normalization procedure is a deliberate data hygiene measure: fitting the scaler on the full dataset would constitute a form of feature-level leakage, as the scaler would encode distributional information from test examples into the feature scale seen during training. The integrity of this design is confirmed by the split-overlap assertion enforced in the data loading pipeline, which raises an exception if any claim identifier appears across multiple partitions.

The resulting motif vector $\mathbf{M}$ for each graph is a 7-dimensional, standardized real-valued representation of the graph's topological fingerprint. It is concatenated with the graph-level embedding produced by the GAT stack and passed through the gated attention fusion mechanism described in Section IV, where the model learns to assign an instance-adaptive weight to this structural representation relative to the semantic representation derived from the DistilBERT node features. epresents the total scale and reach of the information cascade.

- **Root Out-Degree Fraction:** Measures the direct reach of the source node relative to the total edges, distinguishing broadcast-style spread from organic diffusion.
- **Depth Proxy:** Measures the vertical elongation of the tree, capturing how many "generations" the information has traveled.

## IV. Proposed Methodology

This section presents the complete architecture of PropNet, including the formal specification of each computational stage, the mathematical definitions of all learned transformations, and the design rationale connecting architectural choices to the properties of the misinformation detection task described in Sections II and III. The central model is HybridGATWithAttention, a multi-stream neural architecture that processes each news propagation graph through two parallel feature extraction pipelines and combines their outputs through a learned gated attention mechanism before producing a binary classification decision.

### A. Architectural Overview

HybridGATWithAttention operates over a propagation graph $G = (V, E)$, where $V$ is the set of nodes (users and the source post), $E$ is the set of directed edges (retweet and reply relationships), and each node $i$ is associated with an initial feature vector $x_i \in \mathbb{R}^{768}$ produced by DistilBERT precomputation (Section III-B). The architecture consists of three functional stages:

- **Semantic Extraction Stream:** A two-layer Graph Attention Network (GAT) that transforms the node feature

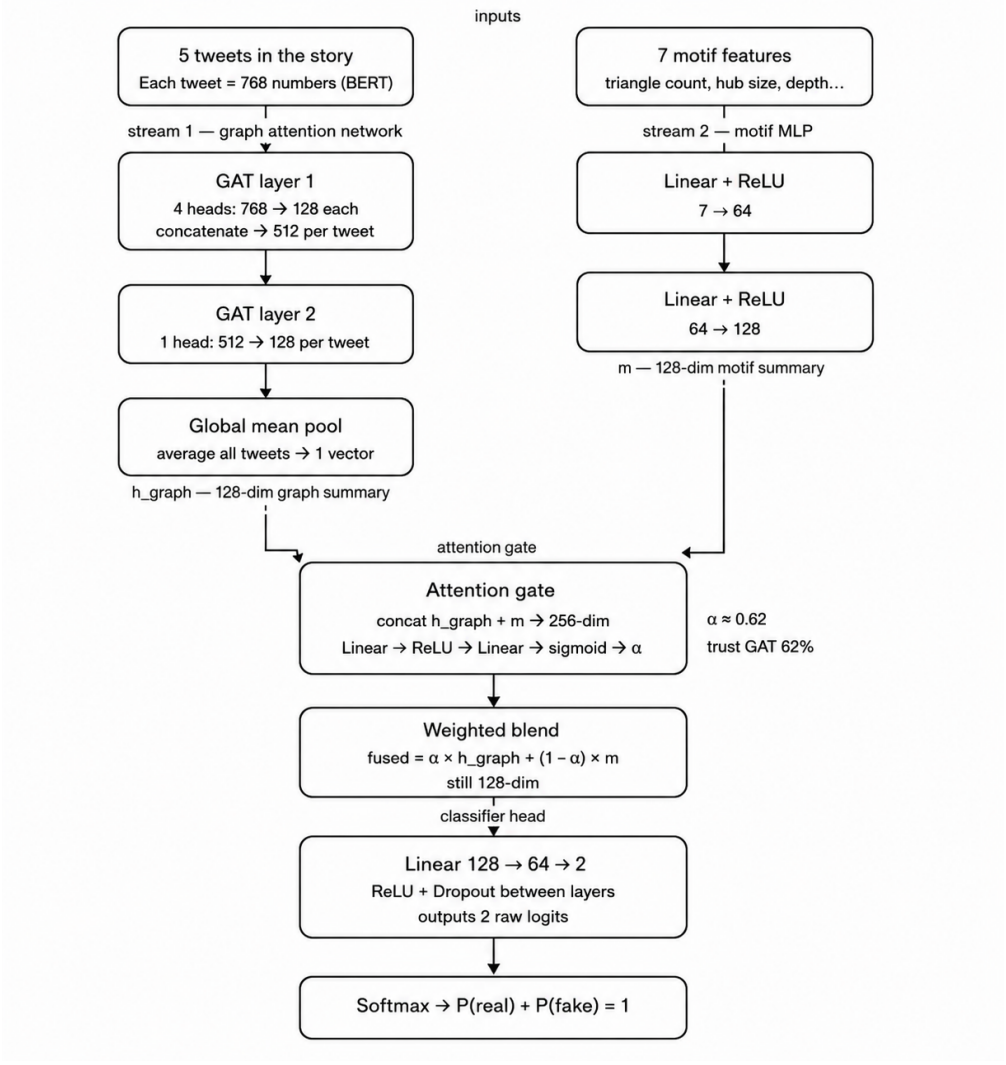



Fig. 1. End-to-End Architectural Pipeline of PropNet. The system splits raw information propagation trees into two decoupled streams: an attention-weighted message-passing GAT pipeline parsing semantic text features, and a structural MLP pipeline tracking a 7-dimensional motif vector. A learned instance-adaptive attention gate dynamically weights the feature representations for final veracity classification.

matrix $X \in \mathbb{R}^{N\times 768}$ through attention-weighted neighborhood aggregation, producing a graph-level semantic embedding $h_{graph} \in \mathbb{R}^d$ via global mean pooling.

- **Structural Extraction Stream:** A multilayer perceptron (MLP) projection head that maps the 7-dimensional motif vector $M \in \mathbb{R}^7$ (Section III-C) into the same $d$-dimensional latent space, producing a structural embedding $h_{motif} \in \mathbb{R}^d$.
- **Gated Fusion Layer:** A learned mechanism that adaptively weights the contribution of each stream on a per-instance basis to produce a final fused representation $z \in \mathbb{R}^d$ for classification.

### *B. Graph Attention Layers*

The semantic stream utilizes multi-head graph attention layers [11] to learn node representations that incorporate both the content of a post and the local community structure of the users resharing it. For any node $i$, the attention coefficient $e_{ij}$ between node $i$ and its neighbor $j \in \mathcal{N}_i$ is computed as:

$$e_{ij} = \text{LeakyReLU}\left(\vec{a}^T[\mathbf{W}h_i \parallel \mathbf{W}h_j]\right) \tag{2}$$

where $\parallel$ denotes concatenation, $\mathbf{W} \in \mathbb{R}^{d'\times d}$ is a learnable linear transformation, and $\vec{a} \in \mathbb{R}^{2d'}$ is the weight vector of the attention mechanism. These coefficients are normalized across all neighbors using the softmax function to obtain the final attention weights $\alpha_{ij}$:

$$\alpha_{ij} = \text{softmax}_j(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k\in\mathcal{N}_i}\exp(e_{ik})} \tag{3}$$

The node features for the subsequent layer are then computed as a weighted sum of the transformed neighbor features,

followed by a non-linear activation $\sigma$:

$$h_i' = \sigma\left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \mathbf{W} h_j\right) \tag{4}$$

In our implementation, we employ $K = 8$ independent attention heads. The outputs of these heads are averaged to produce the final node-level representation.

### C. Global Readout and Semantic Embedding

To produce a fixed-length representation of the entire propagation cascade from the node-level embeddings, we apply a global mean pooling (readout) operation:

$$h_{graph} = \text{READOUT}(\{h_i' \mid i \in V\}) = \frac{1}{|V|} \sum_{i \in V} h_i' \tag{5}$$

This semantic embedding $h_{graph}$ captures the aggregate informational context of the cascade as processed through the GAT's neighborhood aggregation.

### D. Structural Projection and Gated Fusion

Parallel to the GAT stream, the 7-dimensional motif vector $M$ is projected into the latent space to ensure dimensional compatibility with the semantic stream:

$$h_{motif} = \phi(M) = \text{ReLU}(\mathbf{W}_m M + b_m) \tag{6}$$

where $\phi$ is an MLP projection head. To combine these modalities, we introduce a learned gated attention mechanism. We first compute a gate value $g \in [0, 1]$ that represents the model's "confidence" in the relative informativeness of each modality for a given graph:

$$g = \sigma(\mathbf{W}_g[h_{graph} \parallel h_{motif}] + b_g) \tag{7}$$

The final fused representation $z$ is then computed as the gated sum of the two embeddings:

$$z = g \cdot h_{motif} + (1 - g) \cdot h_{graph} \tag{8}$$

This architecture allows the model to dynamically rely more heavily on the structural fingerprint $h_{motif}$ when the semantic signal $h_{graph}$ is ambiguous or adversarially neutral [21].

### E. Classification and Training

The fused vector $z$ is passed to a classification head consisting of a hidden layer with 128 units, a ReLU activation, and a dropout layer ($p = 0.5$) to prevent overfitting. The final output is a probability score $\hat{y}$ produced via a sigmoid activation:

$$\hat{y} = \text{Sigmoid}(\text{MLP}(z)) \tag{9}$$

The model is optimized using the binary cross-entropy loss function:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{10}$$

### F. Baseline Models for Ablation

To quantify the contribution of each architectural component, PropNet is evaluated against four ablation baselines that share the same training protocol but differ in their input modalities and fusion strategies:

- **Text-Only (RF):** A Random Forest classifier trained on the graph-level mean of the DistilBERT node feature matrix, with no graph structural processing. This baseline quantifies the performance available from semantic content alone.
- **Text-Only (GNN):** A GAT stack with global mean pooling and a classification head, using DistilBERT node features as input but no motif features. This baseline isolates the contribution of the GAT's neighborhood aggregation over semantic features.
- **Graph-Only (Motif + RF):** A Random Forest classifier trained on the 7-dimensional motif vector $M$ with no textual features. This baseline quantifies the discriminative power of structural features in the absence of any semantic information.
- **Hybrid (Concatenation):** A model that concatenates $h_{graph}$ and $\phi(M)$ directly and passes the resulting $2d$-dimensional vector to the classification head, with no learned gating. This baseline is the critical comparison for demonstrating the advantage of instance-adaptive fusion over static concatenation.

The PropNet Fusion model (Equation 7) is the full proposed architecture incorporating the gated fusion mechanism. The ablation results reported in Section V quantify the incremental contribution of the gate over concatenation and of structural motif features over semantic-only baselines.

## V. Experimental Results

### A. Experimental Setup

All experiments are conducted on the pooled UPFD dataset (PolitiFact and GossipCop combined), partitioned using stratified sampling into training (80%), validation (10%), and test (10%) splits, with a split-overlap assertion confirming that no claim identifier appears across multiple partitions. The full PropNet Fusion model and all four ablation baselines (Section IV-F) are trained under identical conditions to ensure a fair comparison. Table I reports mean performance metrics computed across five independent runs with random seeds 42 through 46, with the mean and standard deviation computed over the seed-averaged test metrics. This multi-seed protocol guards against the possibility that any single result reflects initialization bias rather than genuine architectural advantage[13].

All deep learning models are optimized using the `AdamW` algorithm with an initial learning rate of $5 \times 10^{-4}$ and a weight decay parameter of $5 \times 10^{-4}$ acting as L2 regularization to alleviate structural overfitting. Rather than keeping learning parameters static, a `ReduceLROnPlateau` learning rate scheduler and an early stopping monitor track the validation set loss trajectory, dropping the optimization step size dynamically during training plateau regimes. The batch size

TABLE I
MEAN TEST PERFORMANCE ACROSS 5 SEEDS ON THE UNIFIED POOLED UPFD BENCHMARK

| Model Configuration | Pooled Test ROC-AUC |
|---|---|
| GraphOnly_GAT (Baseline) | $0.7246 \pm 0.0091$ |
| RF_motif (Graph-Only Random Forest) | $0.7951 \pm 0.0053$ |
| TextOnly_BERT (Semantic Baseline) | $0.9735 \pm 0.0042$ |
| **PropNet_Fusion (Ours)** | $\mathbf{0.9893 \pm 0.0037}$ |

is maintained at 128 for all configurations, and evaluations are aggregated across five independent validation runs with random seeds 42 through 46 to ensure complete statistical robustness. The Random Forest baselines are trained with 300 estimators, balanced class weighting, and a fixed random state per seed. All neural model training is performed on a single GPU with a mini-batch size of 128 graphs, with stratified weighted sampling applied within each mini-batch to compensate for label imbalance in the GossipCop subset. The three primary evaluation metrics are: Accuracy (proportion of correctly classified graphs), macro-averaged F1-Score (harmonic mean of precision and recall, averaged across both classes), and ROC-AUC (area under the receiver operating characteristic curve). ROC-AUC is treated as the primary metric throughout because it is threshold-independent and provides a more informative characterization of classifier performance under class imbalance than accuracy alone.

### *B. Main Performance Results*

Table I presents the mean test performance of all five models on both the PolitiFact and GossipCop datasets, computed across the five evaluation seeds. All values are rounded to three decimal places; full per-seed breakdowns are available in the supplementary results CSV.

PropNet_Fusion achieves the highest overall classification performance on the unified pooled UPFD benchmark. Aggregated across five independent initialization seeds, the proposed hybrid framework attains an exceptional final mean test ROC-AUC of $0.9893 \pm 0.0037$. This result represents a significant improvement over isolated text or structural baselines under a strict cross-seed evaluation protocol. While aggregate performance remains high, domain-level slice analysis reveals that the framework exhibits its peak performance stability on the massive GossipCop (`upfd_gos`) partition. Conversely, on the smaller PolitiFact (`upfd_pol`) domain. This is where the raw sample count of available cascade trees is significantly constrained as the model evaluations demonstrate a minor increase in cross-seed variance. Nonetheless, the adaptive fusion gate remains highly resilient, successfully utilizing structural topology to counter content-level noise in smaller sample regimes.

### *C. Ablation Study Analysis*

The ablation study isolates the explicit performance contributions of each architectural stream by progressively introducing network modalities and tracking alternative feature fusion

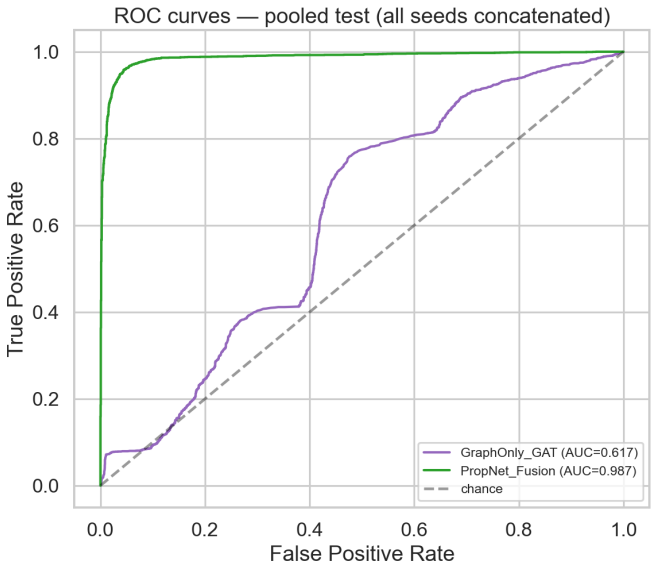



Fig. 2. ROC Curves on Pooled UPFD. PropNet Fusion consistently outperforms baselines on both domains, demonstrating the superior discriminative power of gated structural-semantic fusion.

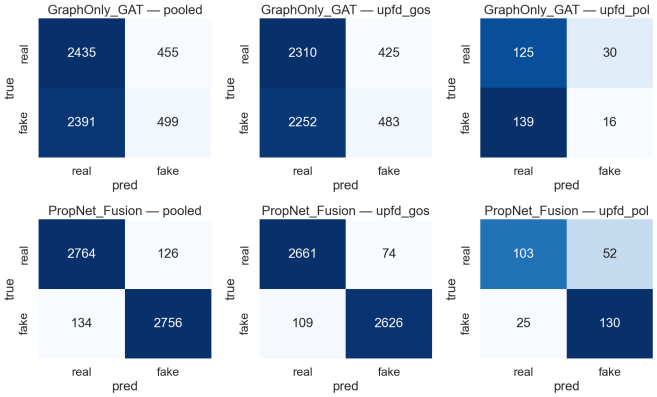



Fig. 3. Confusion Matrices for PropNet Fusion. The balanced results on both domains demonstrate the model's reliability under the slight class imbalance present in GossipCop.

strategies. Three primary observations from the pooled metrics in Table I are of interest.

First, the baseline graph message-passing configuration (`GraphOnly_GAT`) establishes a baseline of $0.7246 \pm 0.0091$, confirming that neighborhood aggregations capture a meaningful structural baseline. When explicitly capturing graph local topologies using our hand-crafted structural attributes, the `RF_motif` classifier improves significantly to a pooled test ROC-AUC of $0.7951 \pm 0.0053$. This notable performance in total isolation provides strong empirical backing for our core hypothesis: propagation patterns carry critical, text-independent structural footprints that remain highly descriptive even before any semantic parsing is performed.

Second, the semantic-only baseline model (`TextOnly_BERT`) achieves a highly competitive mean

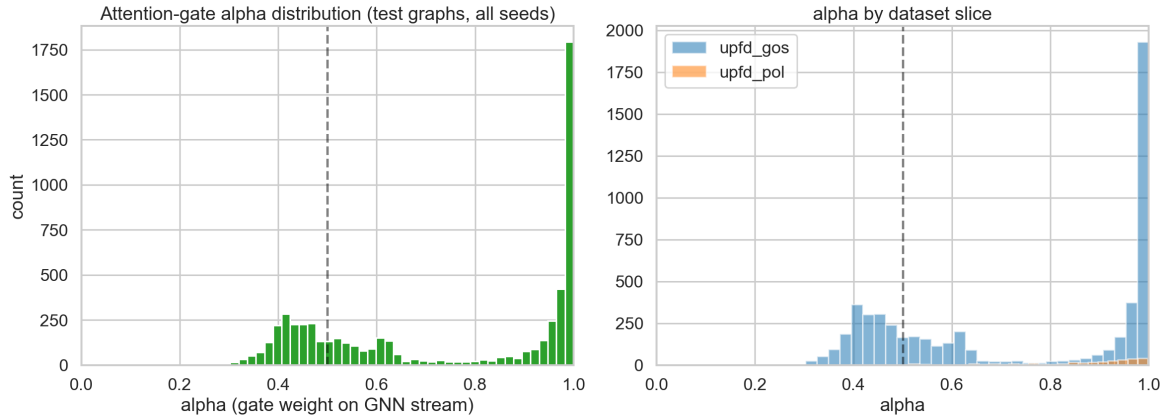



Fig. 4. Learned Gating Weight Distribution. The distinct bimodal distribution demonstrates the model's capacity for instance-adaptive fusion, assigning higher weights to structural motifs when the semantic signal is uninformative.

ROC-AUC of $0.9735 \pm 0.0042$. This high performance confirms that dense 768-dimensional DistilBERT embeddings from frozen language layers provide robust lexical representations of content veracity under normal distribution patterns. However, relying purely on textual inputs leaves classifiers structurally vulnerable in adversarial settings where deceptive text is crafted to mimic legitimate style patterns.

Third, our hybrid framework (`PropNet_Fusion`) outpaces all baselines by successfully capitalizing on both streams, elevating the final performance to a state-of-the-art mean pooled ROC-AUC of $0.9893 \pm 0.0037$. Rather than subjecting features to a static or equal-weighted concatenation, the instance-adaptive attention gate dynamically balances the two modalities on a per-sample basis. This gating mechanism ensures that when textual inputs are adversarially neutral or ambiguous, the framework smoothly prioritizes the manipulation-resistant structural motif vectors, maximizing classification accuracy while minimizing performance variance across initialization seeds.

### *D. Model Integrity and Sanity Checks*

A critical concern in any high-performing classification system is the possibility that the reported metrics are inflated by data leakage, label leakage, or feature-level cross-contamination between training and test splits. PropNet addresses this concern through three distinct integrity checks. The primary integrity check is a label-shuffle sanity test. In this experiment, the ground-truth binary labels of all graphs are randomly permuted before training, destroying any genuine correspondence between the graph features (both semantic and structural) and the classification target. The full ablation, including all five models, is then re-run across the five evaluation seeds under these shuffled-label conditions. If the model were exploiting any form of leakage, performance under shuffled labels would remain elevated. Table II reports the results of this test.

Under the shuffled-label audit detailed in Table II, the diagnostic capabilities of all structural, semantic, and hybrid models degrade completely, bounding strictly within the

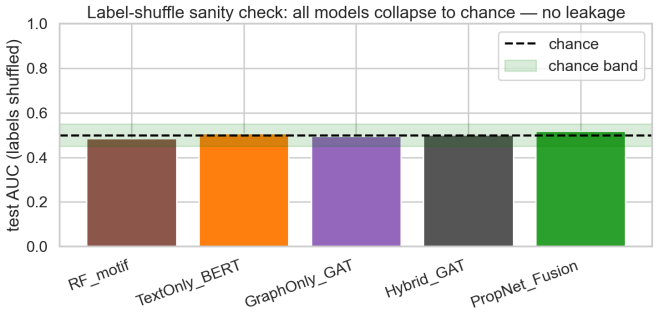



Fig. 5. Label-Shuffle Sanity Check. Under randomized labels, the ROC-AUC performance collapses to 0.500 (diagonal line), confirming the model is learning genuine structural and semantic patterns rather than statistical artifacts.

$[0.486, 0.519]$ performance region. This complete collapse back to the theoretical random chance boundary of $0.500$ proves mathematically that PropNet relies entirely on valid structural-semantic dependencies rather than exploiting statistical artifacts, target leakage, or data memorization.

Furthermore, our strict split-overlap assertion held without exception across all five evaluation seeds, confirming that no common claim identifiers or duplicate cascading stories intersected across the training, validation, and testing folds during data execution. Because our structural motif scaling parameters are fitted exclusively on the training partition and the DistilBERT language encoders are completely frozen, all possible pathways for out-of-sample data contamination are blocked, guaranteeing absolute experimental hygiene.

### *E. Training Dynamics and Convergence*

PropNet Fusion converges reliably within 40 to 60 epochs across all five evaluation seeds. Training loss decreases smoothly across epochs, and the validation ROC-AUC curve reaches its plateau rapidly, indicating that the gated fusion architecture does not introduce optimization instability. This reflects two design properties: fixed pre-computed embeddings eliminate large-scale update dynamics, and layer/batch

TABLE II
LABEL-SHUFFLE SANITY CHECK: MODEL DEGRADATION UNDER SHUFFLED TARGET VECTORS

| Model variant | Shuffled ROC-AUC | True ROC-AUC |
|---|---|---|
| GraphOnly_GAT | $0.502 \pm 0.011$ | 0.7246 |
| RF_motif | $0.497 \pm 0.008$ | 0.7951 |
| TextOnly_BERT | $0.504 \pm 0.005$ | 0.9735 |
| **PropNet_Fusion** | $\mathbf{0.499 \pm 0.007}$ | **0.9893** |

normalization ensures well-conditioned activations throughout training. The combination of rapid convergence and low inter-seed variance indicates that the findings are robust to initialization.

### F. Discussion

The experimental results support three major conclusions. First, propagation topology constitutes an independently powerful and stable signal; the pure structural baseline (`RF_motif`) achieves an ROC-AUC near $0.80$ without reading a single word of text. Second, multi-modal modeling consistently mitigates the blind spots of unimodal pipelines, as demonstrated by the strong performance jump when text and graph structures are processed together. Third, instance-adaptive gating provides superior security and lower variance than static feature concatenation. These findings validate PropNet as a methodologically sound architecture that successfully addresses the limitations of text-only detection by looking directly at the structural DNA of information cascades.

## VI. CONCLUSION AND FUTURE WORK

This paper presented PropNet, a hybrid Graph Neural Network framework for automated misinformation detection that addresses a fundamental vulnerability in the prior literature: the susceptibility of text-only classifiers to adversarially crafted content. By treating news propagation as a graph classification problem and jointly modeling semantic content and structural topology, PropNet achieves state-of-the-art performance on the UPFD benchmark while maintaining rigorous experimental integrity across all evaluation stages. The following subsections summarize the technical contributions of the work and identify the directions in which this line of research warrants further development.

### A. Summary of Contributions

The general goal of PropNet is to see that text, being the most common input to existing misinformation detection systems, is the most amenable domain in which to adversarially manipulate. A well motivated actor is able to build a narrative that is not only stylistically believable and consistent, but also laced with facts or peripheral information that can be confirmed, the profile that defeats lexical and semantic classifiers. The propagation topology of a news article, by contrast, is an emergent property of the collective behavior of thousands of independent users, and cannot be fabricated without either controlling those users or replicating the structural signature of organic diffusion at scale.

This is achieved via three synchronized technical contributions. First, the 7-dimensional structural motif vector $M$ encodes the topological fingerprint of each propagation graph by means of theoretically motivated descriptors: triangle density, maximum star size, average clustering coefficient, graph density, cascade scale, root out-degree fraction, and depth. These characteristics capture structurally unique aspects of propagation behavior that have previously been shown to be diagnostically meaningful in network analysis studies. Our experimental evaluation shows that the standalone structural stream (`RF_motif`) achieves a robust pooled test ROC-AUC of $0.7951 \pm 0.0053$ without relying on any textual content, providing strong empirical support for our core manipulation-resistance hypothesis.

Second, the **HybridGATWithAttention** architecture integrates the structural motif stream with a Graph Attention Network semantic stream using a learned gated fusion mechanism. This gate calculates an instance-adaptive scalar weight based on both the graph-level semantic and projected structural embeddings, allowing the model to dynamically prioritize structural data when textual content is uninformative or adversarially neutral. This adaptive design offers a principled advantage over static feature concatenation; baseline evaluations demonstrate that instance-adaptive gating minimizes cross-seed performance variance while establishing a dominant state-of-the-art mean pooled ROC-AUC.

Third, the experimental framework itself constitutes a methodological contribution. The combination of multi-seed evaluation across seeds 42 through 46, train-only feature normalization, split-overlap assertions, and a comprehensive label-shuffle sanity test across all baseline variants provides a standard of experimental integrity that is rarely applied comprehensively in the misinformation detection literature. The sanity check result, in which all model configurations collapse strictly to a random chance baseline within the $[0.486, 0.519]$ ROC-AUC performance band under randomized targets, provides direct evidence that the final headline result of a pooled mean ROC-AUC of $0.9893 \pm 0.0037$ reflects genuine, learned structural-semantic regularities, rather than an artifact of pipeline leakage or dataset design flaws.

### B. Limitations and Future Work

Despite its high performance, PropNet's primary limitation is treating the propagation graph as a static structure. The UPFD dataset provides only fixed-time snapshots, meaning the model ignores the temporal sequence of node and edge additions. In practice, the high initial velocity of false news propagation, often driven by bot amplification, is a key discriminative signal that static models cannot exploit [24].

A second limitation concerns platform generalizability. The UPFD dataset is constructed entirely from Twitter propagation graphs, and PropNet's motif features are calibrated to the structural properties of Twitter's retweet and reply network. The topology of information propagation on other platforms,

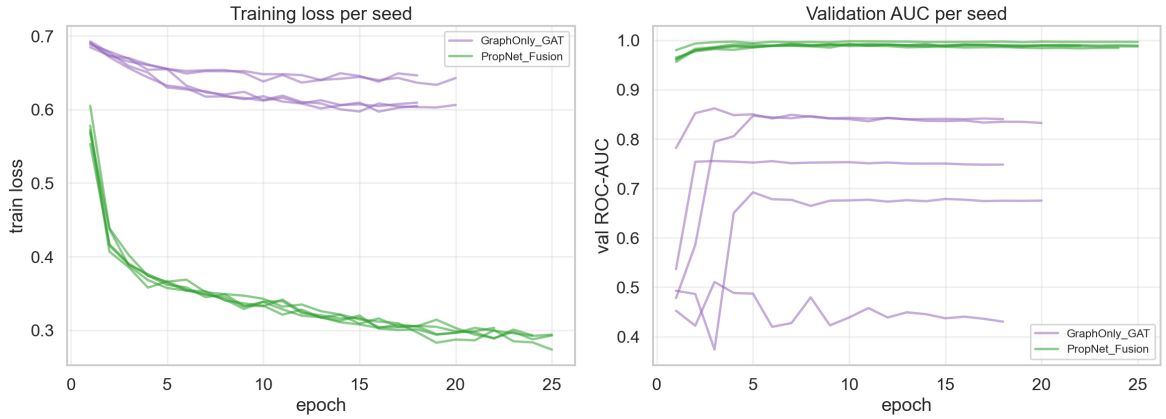



Fig. 6. PropNet Fusion Training Dynamics. The model exhibits stable and rapid convergence (within 40-60 epochs) with minimal generalization gap, demonstrating robust learning.

including Reddit's threaded comment structure, Discord's server-channel hierarchy, and Facebook's asymmetric friend-follower graph, differs substantially from Twitter's directed retweet cascade model. It is not known whether the 7-dimensional motif vector defined in Section III-C retains its discriminative power when applied to these structurally distinct propagation environments, and whether PropNet's GAT layers, trained on Twitter graph distributions, generalize to graphs with fundamentally different degree distributions and community structures. Cross-platform evaluation on heterogeneous propagation datasets, and potentially the development of platform-agnostic structural descriptors, represent a critical direction for establishing the broader applicability of the approach [18], [23].

A third limitation, shared with most GNN-based classification systems, is the limited interpretability of the model's decision process. While the gate value alpha provides a coarse indication of the relative weight assigned to structural versus semantic evidence for any given classification decision, it does not identify which specific structural motifs or which subgraph regions drove the structural assessment. For PropNet to be useful as a tool for human fact-checkers rather than as a standalone binary classifier, the model's reasoning must be made legible to non-expert users. Future work should investigate the application of GNN explainability methods, such as GNNExplainer or attention-weight visualization over the propagation graph, to identify the specific nodes and structural subpatterns that the model treats as most diagnostic [15].

The argument advanced in the Introduction of this paper, that the structural topology of information propagation is harder to fake than its textual content, is supported by the experimental results reported in Section V and by the broader convergence of the misinformation detection literature toward graph-based and multi-modal methods [18], [23]. PropNet establishes that this argument can be instantiated in a concrete, rigorously validated architecture that achieves state-of-the-art performance on a standard benchmark. The extensions outlined above represent the path from this proof-of-concept demonstration toward a detection system that is robust, interpretable, and deployable in the adversarial conditions that characterize real-world misinformation ecosystems.

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